



OPEN Spatial pattern characteristics and influencing factors of state-level sports industry bases in China

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The spatial distribution of sports industry bases in China holds significant practical importance for the high-quality development of China's sports industry. This study employs spatial data analysis (ESDA), the spatial Gini coefficient, the imbalance index, the standard deviation ellipse, and a panel negative binomial regression model to examine the spatial patterns and influencing factors of the distribution of State-level Sports Industry Bases. The findings are as follows: (1) The 334 State-level Sports Industry Bases in China are located primarily east of the "Hu Line." The distribution pattern of these bases across the 31 provinces has undergone continuous changes from 2016 to 2021. (2) The overall distribution of State-level Sports Industry Bases in China's seven major geographic regions is uneven. Northeast China has the most balanced distribution, while East China has the highest concentration. The standard ellipse of the distribution from 2016 to 2021 reveals a trend of "discrete-centered-discrete-centered-centered." (3) Economic growth and government financial investment in each province and region are significantly positively correlated with an increase in the number of State-level Sports Industry Bases. In contrast, higher levels of regional innovation inhibit the growth of these bases.

Keywords Sports industry, Sports industry bases, Spatial pattern, Panel negative binomial model

Sports industry bases constitute an important form of deepening the reform in the field of sports industry and promoting the optimization and upgrading of the structure of the sports industry in China. In 2011, the State General Administration of Sports issued the "Measures for the Administration of State-level Sports Industry Bases (for Trial Implementation)", which defines State-level Sports Industry Bases as an area named by the State Sports General Administration as having a considerable foundation, scale and characteristics in the development of the sports industry, or an organization that has significant influence and strong competitiveness in a certain field of the sports industry. In March 2016, on the basis of recent work experience, the General Administration of Sport of China revised the *Administrative Measures for State-level Sports Industry Bases (Trial Implementation)* and formally issued the *Circular on Strengthening the Construction of State-level Sports Industry Bases*, which expands and clarifies the concepts and content system of State-level Sports Industry Bases. The "state-level sports industry bases" before 2015 were taken as the general term for this work, and three types were set up, namely, state-level sports industry demonstration bases, state-level sports industry demonstration units, and state-level sports industry demonstration projects. As of 2021, the 334 State-level Sports Industry Bases named and recognized by the State General Administration of Sports include 79 demonstration bases, 143 demonstration units and 112 demonstration projects, covering 31 provinces and 5 cities with separate plans.

Relevant studies have shown that the behavior of industrial agglomeration increases interfirm linkages, enhances industrial concentration^{1,2}, promotes the economic profitability of the industry, allows for the concentration of the efforts of economic entities to create added value, creates new conditions for the development of both traditional and emerging sectors of the economy, and enhances the resilience of industry^{3,4}. Sports industry agglomeration helps promote the high-quality development of green energy⁵ as well as realize the green and low-carbon development of the sports industry⁶; at the same time, it helps determine the functional positioning of the city, enhances the brand image and attractiveness of the city, increases the attention of residents to sports, and contributes to the improvement of the overall health of society⁷. As a special manifestation of sports industry agglomeration, the formation of sports industry bases further drives the development of neighboring commercial institutions⁸. However, although state-level sports industry bases have had good development momentum in recent years, real problems such as a small stock scale, insufficient innovation drive, and unbalanced regional layout simultaneously exist⁹. In addition to China's vast territory, the

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socioeconomic levels of regions differ greatly, and the development conditions and development levels of state-level sports industry bases clearly differ across regions¹⁰.

On this basis, what characteristics does the distribution of State-level Sports Industry Bases present in the analysis? What are the socioeconomic factors affecting their distribution? Clarifying the above issues is highly practical for the high-quality development of China's sports industry. Therefore, the marginal contributions of this paper are as follows: (1) take the state-level sports industry base as an emerging sports economic geographical unit and explore the reality of the development of the state-level sports industry base in China from spatial and temporal dimensions. (2) To systematically grasp the driving factors of the spatial layout characteristics of state-level sports industry bases and to propose targeted optimization paths to provide reference bases for the formulation of relevant policies and the career development of relevant practitioners and to promote the high-quality development of the regional sports industry.

Literature review

Research on the development of sports industry bases in China

The concept of an industrial base was first proposed by Hopland, the president of the American DEC, in 1984; Hopland believed that an industrial base is one or several related product-intensive, R&D-intensive, talent-intensive and enterprise-intensive areas¹¹. Sports industry clusters around the world exhibit diverse characteristics. In developed countries, sports industry clusters are typically formed spontaneously by market forces, such as the MotoSport Valley racing cluster in the UK and the Montebelluna sports footwear cluster in Italy¹²; the surfing industry cluster in France; the sailing industry cluster in New Zealand¹³; and the ocean racing cluster in Brittany, France¹⁴. However, there are also clusters established with government support, such as the e-sports industry cluster in Alamein, Egypt¹⁵. The sports industry bases in China are developed as part of a long-term economic strategy aimed at ensuring national and regional economic stability. Under the guidance of relevant government departments, these bases are designed as open projects with the goal of enhancing comprehensive strength. They are intended to address the inadequacies and failures of market mechanisms, with the evaluation and construction processes being led by the government.

Currently, scholars' attention to the sports industry focuses primarily on three areas: the development strategies of sports industry bases, the development models of the sports industry bases in specific regions, and the evaluation of the competitiveness of these bases. (1) First, the sustainable development of sports industry bases requires strengthening the measurement of the aggregation level of these bases, improving management methods, and advancing dynamic regulatory measures. It also necessitates enhancing benchmarking, promoting transformation and upgrading, achieving coordinated cooperation, and strengthening market regulation⁹. (2) Second, further attention is given to the development models of sports industry bases in different regions. Yaroslav¹⁶ demonstrated the interactive mechanism within the "National-Society-Science-Enterprise" system and established an organizational and structural model for sports industry clusters. This mechanism enhances the efficiency of joint activities among economic entities in the sports industry. Zhao et al.¹⁷ suggested that the sports service industry could be the core industry for building a sports industry base in Guangzhou, implementing a planning layout based on the "One Belt, One Network, and Three Major Areas" strategy. Furthermore, strengthening government regulation¹⁸, enhancing smart manufacturing capabilities¹⁹, and fostering cooperation between research institutions and enterprises to establish an industrial big data service platform²⁰ are crucial drivers of the development of sports industry bases. (3) Moreover, based on a clear understanding of the development models and strategies of the sports industry, how can the competitiveness of these bases be assessed? Zhao²¹ utilized the "diamond model," which summarized four dimensions: factor conditions, industry conditions, demand conditions, and government conditions. Zhang²² applied the fuzzy comprehensive evaluation method and constructed a competitiveness evaluation index system for China's sports industry bases by selecting dimensions such as technological innovation capability, product market competitiveness, and corporate cultural innovation ability.

Research on the layout and influencing factors of sports industry bases

Locational advantage is a fundamental guarantee for achieving high-quality industrial development. The geographical location of industrial bases is closely linked to the development resources available to them. Consequently, the rationality of the industrial base distribution and its influencing factors have increasingly become topics of scholarly debate. Research indicates that the spatial distribution of sports industry bases in China aligns closely with the three major strategic regions of the country—the Beijing–Tianjin–Hebei collaborative development, the Yangtze River Economic Belt, and the Belt and Road Initiative. This distribution is evolving toward a more multilevel and three-dimensional framework¹⁰. The core aggregation areas of China's State-level Sports Industry Bases are centered primarily around key urban clusters, forming peripheral radiation zones and extended corridors. These areas exhibit distinct core diffusion characteristics but also display relatively dispersed spatial distributions, with relatively high densities in the eastern and northern coastal regions²³.

Further analysis of the factors influencing the distribution of sports industry bases reveals that economic level, market conditions, technological investment, the consumer market, labor force elements, and socioeconomic proximity are all considered critical influencing factors^{24,25}. In specific case studies, policy coordination and stakeholders play a significant role in the development of the esports industry cluster in Egypt²⁶. Viljamaa's research on the UK motorsports cluster revealed that knowledge creation and transfer are key success factors in maintaining competitive advantage²⁷. Gerke's study on the Auckland sailing cluster in New Zealand and the Victoria surfing cluster in Australia revealed that, compared with geographical proximity, interorganizational innovation links have a more substantial impact on the sustainable development of industry clusters²⁸. The layout of industrial bases is clearly influenced by a variety of factors. However, limited attention has been given to the factors influencing the spatial distribution of sports industry bases in China. As the sports industry is poised

to become a sunrise industry in China, exploring the distribution and influencing factors of China's State-level Sports Industry Bases is highly important for the growth and expansion of the sector.

In summary, several of the studies referenced above are somewhat outdated, with incomplete statistics on China's State-level Sports Industry Bases and fail to reflect the current overall spatial distribution. In the analysis of influencing factors, most studies rely on logical reasoning and lack quantitative research on the relevant factors. The proposed development strategies are not supported by accurate data. The few studies that employ the geographical detector method have several limitations, as this approach only explores the explanatory power (driving force) of influencing factors but cannot determine the direction (positive or negative) of their effects. Furthermore, while studies on the sports industry bases in specific regions are relatively precise, they lack universal evaluation and development models. Therefore, as China's sports industry has entered the 14th Five-Year Plan period, investigating the provincial distribution of more than 300 State-level Sports Industry Bases in China and selecting more scientific analytical methods to explore the impact dimensions and effects of socioeconomic factors on the number of State-level Sports Industry Bases in China are of significant theoretical and practical importance for planning the future development of China's sports industry and its high-quality growth.

Research design

Data collection

This study selects 334 China's State-level Sports Industry Bases recognized by the General Administration of Sport of China from 20,062,021 as the subjects of investigation. The latitude and longitude of different types of State-level Sports Industry Bases were obtained by selecting the coordinate system through the Baidu map. The coordinates of the State-level sports industry demonstration bases are based on the county (city and district) government sites; the coordinates of the State-level sports industry demonstration units and training industry demonstration projects are based on the company registered places, and the coordinates of the State-level sports industry demonstration units with branch offices are based on the company headquarters; the coordinates of the national sports demonstration projects of the sports event type are based on the event hosting places; and the coordinates of the sports stadium type are based on the venues' locations. For sporting event-type state-level sports industry demonstration projects, the place where the event is held is taken as the coordinate. In this way, a vector database of State-level Sports Industry Bases was constructed from 20,062,021, which serves as the basic base map for the information collection and spatial analysis of this study. ArcGIS was used to align and vectorize the study area. Data on the geographical location of State-level Sports Industry Bases and related influencing factors are obtained from the official website of the General Administration of Sports of China (<https://www.sport.gov.cn/>), *China Statistical Yearbook* and IncoPat Patent Database, and the data were processed via Stata 17.0.

Research methods

Exploratory spatial data analysis (ESDA)

Global spatial autocorrelation test

Global spatial autocorrelation measures the degree of spatial correlation and spatial difference in the target region as a whole. The larger the absolute value of the Moran index is, the stronger the spatial correlation. The mathematical formula is defined as:

$$I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\left(\sum_{i=1}^n \sum_{j=1}^n w_{ij} \right) \left(\sum_{i=1}^n (x_i - \bar{x})^2 \right)} \quad (1)$$

where n denotes the total number of regional spatial units, x_i and x_j denote the attribute values of random variable x on geographic units i and j , $\bar{x}$ is the average value of x , and w_{ij} is the spatial weight matrix. To ensure the robustness of the empirical results, we construct the adjacency matrix (w) on the basis of whether the geographic space of each province and domain is adjacent or not; the spatial weight matrix of geographic distance on the basis of the inverse of the straight line distance between provinces and domains (d); and the spatial weight matrix of economic distance on the basis of the inverse of the difference in the GPD of each province and domain (e) to measure the Moran index.

Local spatial autocorrelation test

The local Moran index, which represents the distributional characteristics exhibited by variables in a local subspace system, can further reveal the magnitude of each regional cell's contribution to global spatial autocorrelation and the extent to which the global assessment of spatial autocorrelation masks local instability²⁹. This study was analyzed using the local indicator of spatial association (LISA) index and Moran scatterplot, and the mathematical formula is defined as follows:

$$I_i = \frac{(x_i - \bar{x})}{S^2} \sum_j w_{ij} (x_j - \bar{x}) \quad (2)$$

where S^2 is the discrete variance of x_i and w_{ij} is the weight matrix. A positive value of this index measure indicates that the region has a positive correlation with the surrounding region, and a negative value indicates a negative correlation with the surrounding region.

Standard deviation ellipse

The distribution of the State-level Sports Industry Bases in each province and region presents different characteristics over time, and the specific transfer direction is measured by the standard deviation ellipse with the change in time, which can reflect the transfer of the center of gravity of the distribution of the industry base. The formula for calculating the center of gravity coordinates can be used to obtain the spatial transfer index, and the spatial center of gravity transfer distance formula can be used to obtain the trend direction of the spatial distribution of the national sports industrial base. The formula for calculating the center of gravity coordinates is as follows:

$$\begin{aligned} X_t &= \frac{\sum_{i=1}^n (C_{ti} * X_i)}{\sum_{i=1}^n C_{ti}} \\ Y_t &= \frac{\sum_{i=1}^n (C_{ti} * Y_i)}{\sum_{i=1}^n C_{ti}} \end{aligned} \quad (3)$$

In the above equation, C_{ti} is the area of the city at time i ; the spatial mean of the geometric center of the region is (X_p, Y_p) . When the spatial mean of a region is significantly different from the geometric center of the region, the spatial phenomenon of this parameter is unevenly distributed, i.e., "center of gravity deviation" occurs. The distance of deviation is expressed in terms of the spatial center of gravity shift distance and is calculated by the following formula:

$$D = \sqrt{(M_i - M_j)^2 + (N_i + N_j)^2} \quad (4)$$

Negative binomial regression analysis

Given that the number of national sports industry bases is discrete nonnegative integer data and does not follow a normal distribution upon testing, a count model is more suitable than a linear model for econometric analysis³⁰. A choice is made between the Poisson regression model and the negative binomial regression model within the count models. The calculations reveal that the variance in the number of sports industry bases is much greater than the mean, thus failing to meet the prerequisites for using the Poisson regression model. Further testing of the α (overdispersion parameter) statistic indicates that the negative binomial regression model is more appropriate. However, owing to the presence of a significant number of zero values in the national sports industry bases between 2016 and 2021 across various provinces, which may lead to the "zero-inflation" phenomenon in count data, a choice must be made between the standard and zero-inflated negative binomial regression models³¹. A comparison of the relative sizes of the AIC and BIC statistics confirmed that the standard negative binomial regression model should be selected. Considering the data used in this study, panel standard negative binomial regression analysis is ultimately conducted, as expressed in the following equation:

Let y_i obey the negative binomial distribution, and the probability density function of y_i is

$$f(y_i) = \frac{\Gamma(y_i + \alpha)}{y_i! \Gamma(\alpha)} \left(\frac{\alpha}{\alpha + \theta_i} \right)^\alpha \left(\frac{\theta}{\alpha + \theta_i} \right)^{y_i} \quad (5)$$

where α is the dispersion parameter. To consider the relationship between the independent and dependent variables in the count data, the covariates x in the θ part of the Poisson parameter are introduced, and the panel negative binomial regression model is obtained, which can be expressed as

$$\log(\theta_i) = x_i^T \beta \quad (6)$$

Empirical results and discussion

Spatial distribution of state-level sports industry bases in China

Quantitative distribution of state-level sports industry bases

In terms of regional distribution and the number of recognized bases, the 334 China State-level Sports Industry Bases that have been named and recognized cover 31 provincial regions. In terms of the distribution of the number of Sports Industry Bases, the top five provinces in the country are Zhejiang Province, Jiangsu Province, Fujian Province, Shanghai Municipality and Shandong Province, with 40, 32, 30, 25 and 25 bases, respectively. According to the classification of "three types", among the existing 79 demonstration bases, the top three in terms of quantity are Zhejiang Province (12), Shandong Province (8), and Jiangsu Province (7). Among the 143 demonstration units, Fujian Province (17), Beijing Municipality (16), and Jiangsu Province (15) ranked in the top 3. Among 112 demonstration projects, Zhejiang Province (15), Fujian Province (11), and Shanghai Municipality (10) are among the top 3 in the country (Fig. 1). The construction of Sports Industry Bases in Beijing, Jiangsu, Zhejiang, Shanghai, Fujian and other regions is at a leading level. With the deepening of the concept of industrial integration in recent years, the sports industry has been exchanging and interoperating with many fields. At present, the State-level Sports Industry Bases have formed the trend of the integration and development of sports competition performance, fitness and leisure, sporting goods, stadiums and other industries. Among them, the integration of the sports industry with tourism, culture, recreation, science and technology has become a new growth point for local economic development.

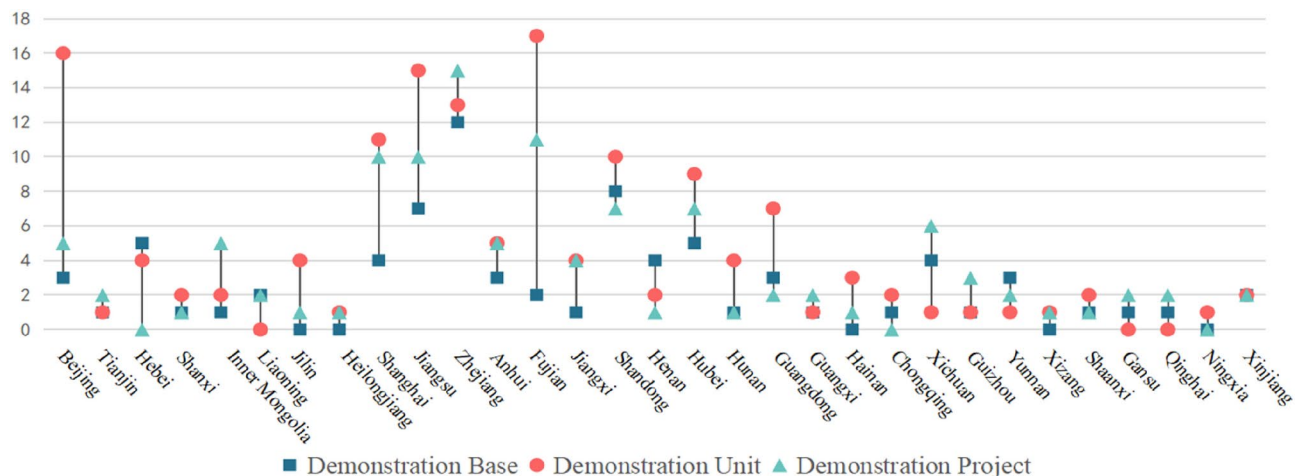


Fig. 1. Provincial distribution of State-level Sports Industry Bases (2006–2021).

Location characteristics of state-level sports industry bases

In 2006, approved by the State General Administration of Sports, Shenzhen set up the first State-level Sports Industry Bases, marking the official start of the exploration and practice of the State-level Sports Industry Bases. From 2006 to 2010, five cities and counties, such as Wenjiang in Chengdu, Jinjiang in Fujian, Longtan Lake in Beijing, Fuyang in Hangzhou, and Leling in Shandong, were approved as State-level Sports Industry Bases. In 2013, according to the "Management Measures of State-level Sports Industry Bases (Trial)", the State General Administration of Sports officially named the first county cluster-type State-level Sports Industry Bases—Jiangsu Province Sunan (county) State-level Sports Industry Bases. In the same year, Beijing Olympic Park was recognized as the first state-level sports industry demonstration unit. In 2016, to fully implement the spirit of Document No. 46 of the State Council, the State General Administration of Sports proposed in the "13th Five-Year Plan for Sports Development" that, by 2020, 50 State-level Sports Industry demonstration bases in counties with larger scale of industry and obvious clustering effects would be set up nationwide, and 100 State-level Sports Industry demonstration units and 100 State-level Sports Industry demonstration projects with high reputation and international influence and strong market competitiveness would be established. To facilitate differentiation through location markers, Fig. 2 displays the locational characteristics of State-level Sports Industry bases for each year from 2016 to 2021, following the standardization of the concept and content system of State-level Sports Industry bases. The distribution density of State-level Sports Industry bases gradually increases from the western region to the central region and then to the eastern region. To further analyze the spatial associations, hotspots, or aggregation areas, as well as the shift in distribution centers of State-level Sports Industry Bases in recent years, exploratory spatial data analysis (ESDA) and spatial center-of-gravity shift models have been employed for in-depth investigations.

Spatial pattern of the distribution of state-level sports industry bases

Global spatial autocorrelation test

Owing to differences in the concept and content system of State-level Sports Industry Bases in China before and after 2016 and given that the majority of provinces had zero State-level Sports Industry Bases prior to 2016 (with only 25 State-level Sports Industry Bases established in China between 2006 and 2015), a sample of 309 State-level Sports Industry Bases from 2016 to 2021 is selected for analysis of spatial distribution patterns and influencing factors. As shown in Table 1, under the three spatial weight matrices, the Moran index of the number of Sports Industry Bases in China's provinces and regions from 2016 to 2018 is mostly positive, and the P value passes the 5% significance test, which is significantly different. The growth of the Moran index in successive years indicates that the state of agglomeration is constantly increasing. Among them, there are strong spatial agglomeration characteristics under the economic distance matrix for three consecutive years, which reflects that the economic strength of each province and region can enhance the agglomeration degree and development scale of the State-level Sports Industry Bases, and it is an important reference factor for the relevant departments to identify the Sports Industry Bases. In 2019, State-level Sports Industry Bases exhibited a random distribution. From 2020–2021, under the neighborhood matrix and geographic distance matrix, State-level Sports Industry Bases differ significantly and present spatial agglomeration characteristics, but the agglomeration state weakens annually. This result shows that, in recent years, the geographical distribution of State-level Sports Industry Bases has become more reasonable, the development mode has become more diversified, and the relevant departments have paid more attention to the balanced development of geographical regions rather than concentrating on economically developed regions in their audit and recognition. Taking Xinjiang as an example, among its only four Sports Industry Bases, three, namely, the Altay State-level Sports Industry Demonstration Base (Demonstration Base), Xinjiang Kerala Investment Company Limited (Demonstration Project), and A-Man Fitness Chain Fitness Project (Demonstration Project), are recognized from 2020 to 2021. However, at the same

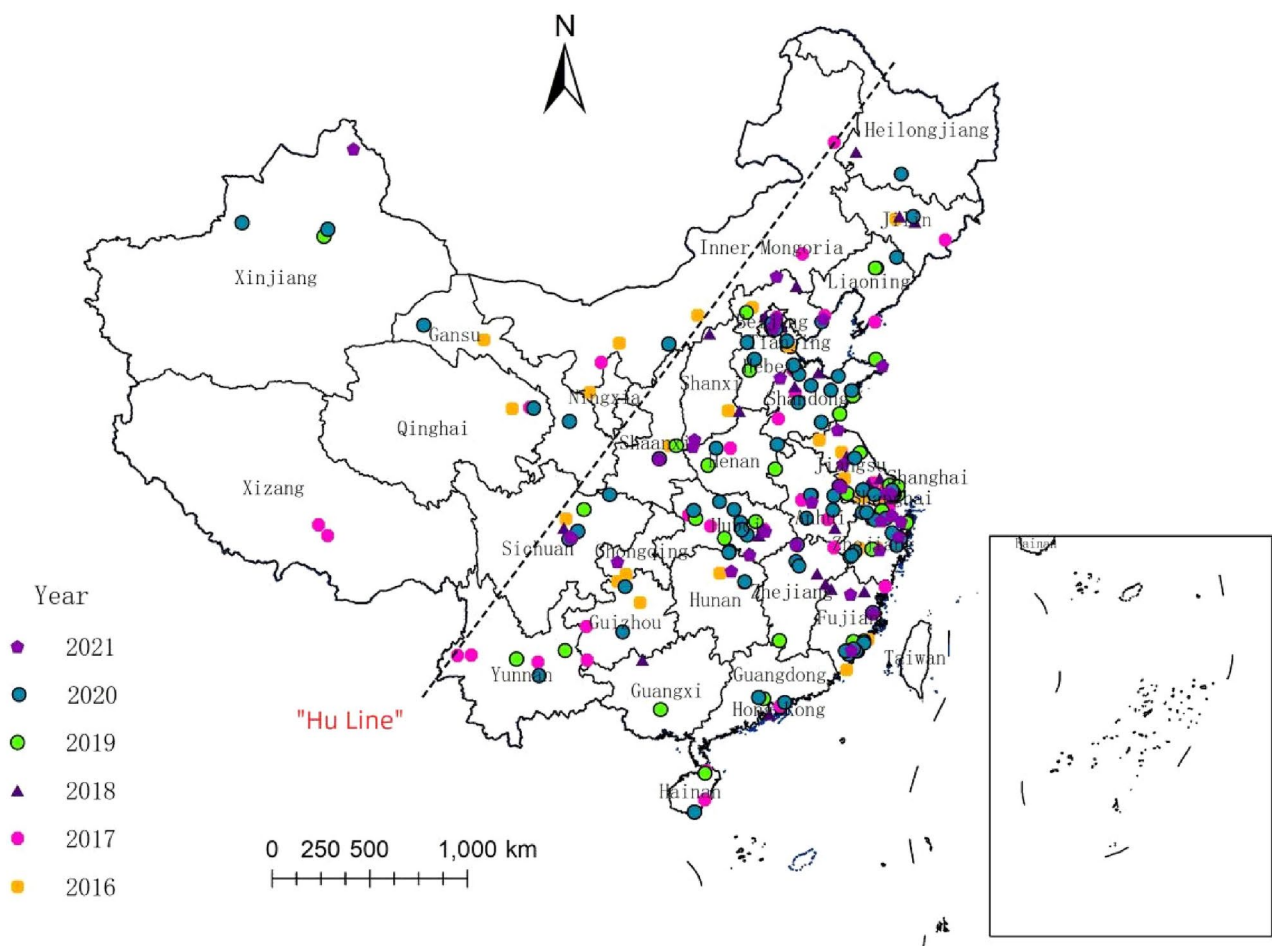


Fig. 2. Time-sharing of the spatial distribution of State-level Sports Industry Bases. *Note:* This figure is based on the standard map No. GS(2020)4619 downloaded from the website of the Standard Map Service of the Ministry of Natural Resources (<https://www.mnr.gov.cn/>), and the base map has not been modified. This figure was produced using the Arc Geographic Information System, ArcGIS, version 10.6 (<https://www.arcgis.com/>).

Year	Adjacency matrix(<i>w</i>)		Geographic distance matrix(<i>d</i>)		Economic distance matrix(<i>e</i>)	
	Moran's <i>I</i>	<i>P</i> -value	Moran's <i>I</i>	<i>P</i> -value	Moran's <i>I</i>	<i>P</i> -value
2016	0.180**	0.040	-0.005	0.359	0.112*	0.067
2017	0.157	0.110	0.041**	0.031	0.140**	0.057
2018	0.483***	0.000	0.131***	0.000	0.174**	0.018
2019	0.150	0.115	0.009	0.212	0.113	0.101
2020	0.484***	0.000	0.155***	0.000	-0.049	0.862
2021	0.326***	0.002	0.057***	0.008	0.091	0.165

Table 1. Moran's *I* test results (2016–2021). *, **, *** denote significance tests at the 10 per cent, 5 per cent and 1 per cent levels, respectively.

time, the dependence of the distribution of State-level Sports Industry Bases on geographical proximity weakens annually.

Local spatial autocorrelation test

The local Moran index indicates the distribution characteristics of the variables exhibited in the local subspace system, 2016 and 2021 were selected as the representative years to generate the Moran scatter plot (Figs. 3, 4 and 5). As shown in Fig. 4, most of the provinces are distributed in the third quadrant, i.e., the low-low agglomeration blind spot area (LL area). Among them, Shanghai, Shandong, Zhejiang and Jiangsu are always in the first quadrant, i.e., the high-high agglomeration hotspot zone (HH zone); these four provinces are located in the eastern coastal region, with a more

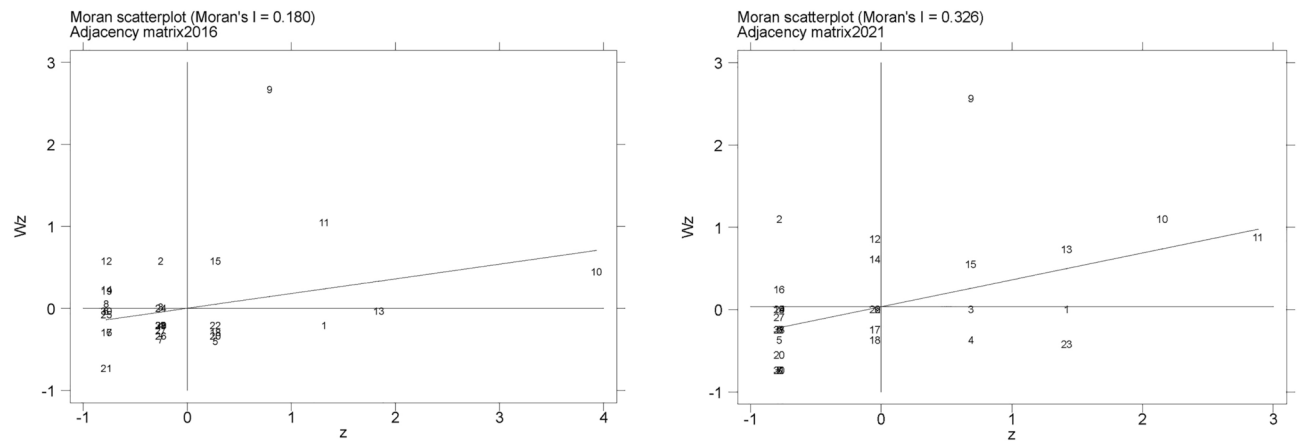


Fig. 3. Moran scatter diagram under adjacency matrix.

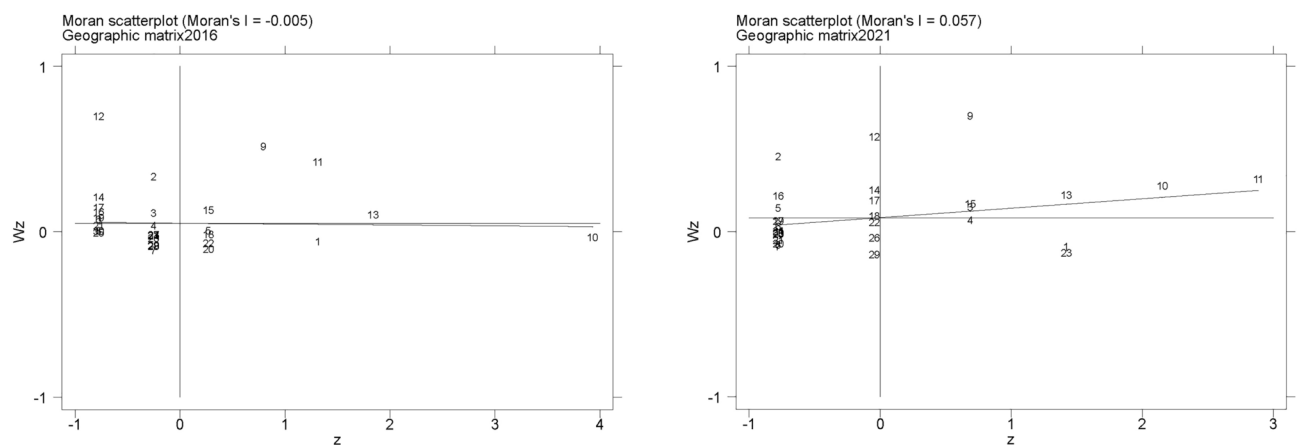


Fig. 4. Moran scatter diagram under geographic matrix.

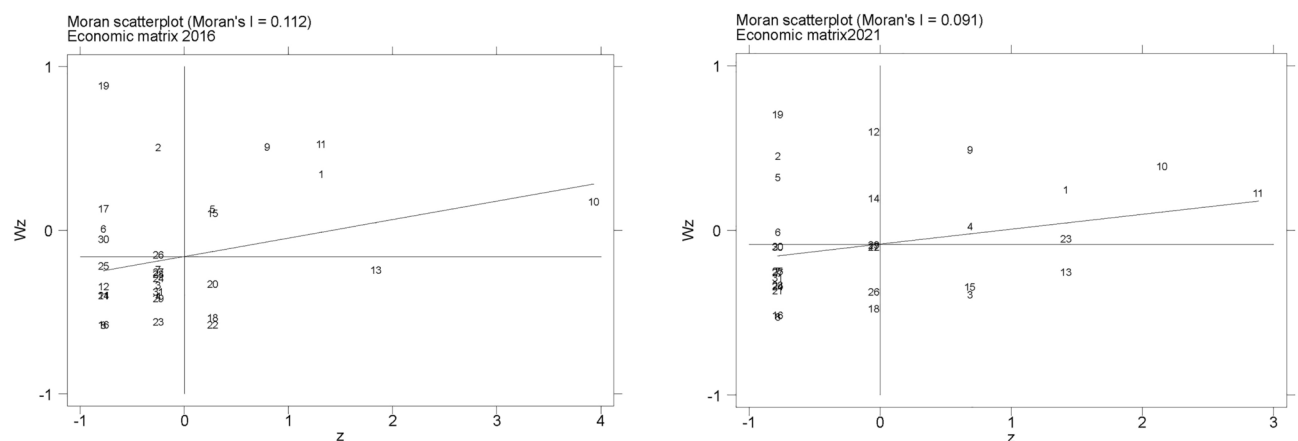


Fig. 5. Moran scatter diagram under economic distance matrix. *Note:* No. 1 for Beijing, No. 2 for Tianjin, No. 3 for Hebei, No. 4 for Shanxi, No. 5 for Inner Mongolia, No. 6 for Liaoning, No. 7 for Jilin, No. 8 for Heilongjiang, No. 9 for Shanghai, No. 10 for Jiangsu, No. 11 for Zhejiang, No. 12 for Anhui, No. 13 for Fujian, No. 14 for Jiangxi, No. 15 for Shandong, No. 16 for Henan, No. 17 for Hubei, No. 18 for Hunan, No. 19 for Guangdong, No. 20 for Guangxi, No. 21 for Hainan, No. 22 for Chongqing, No. 23 for Sichuan, No. 24 for Guizhou, No. 25 for Yunnan, No. 26 for Xizang, No. 27 for Shaanxi, No. 28 for Gansu, No. 29 for Qinghai, No. 30 for Ningxia, No. 31 for Xinjiang.

developed economy, stronger consumption ability of the residents, and better development momentum of the sports industry. Tianjin, Anhui and Jiangxi are always in the second quadrant, i.e., the low–high–strong enemy surrounding area (LH area), and the number of Sports Industry Bases relatively lags behind the neighboring regions; Beijing and Shanxi are always in the fourth quadrant, i.e., the high–low–crane flocking area (HL area). This shows that the number of Sports Industry Bases is relatively ahead of the neighboring regions. Shanxi enters the HL zone from the LL zone, i.e., through its efforts, the number of Sports Industry Bases surpasses that of neighboring provinces. Chongqing enters the LL zone from the HL zone, i.e., the number of its own Sports Industry Bases declines, and Fujian enters the HH zone from the LH zone, indicating that the number of Sports Industry Bases gradually catches up with that of neighboring provinces. Under the geographical weight matrix, most provinces are distributed in the second and third quadrants. The southeastern coastal provinces are located in the LL zone. Its number of Sports Industry Bases in 2016 showed a negative spatial correlation, i.e., provinces with a high number of Sports Industry Bases were mostly surrounded by provinces with a low number, whereas provinces with a low number of bases were surrounded by a greater number of provinces with a high number of bases. In the regional quadrant change, Inner Mongolia enters the HL zone from the LH zone, i.e., the number of its own Sports Industry Bases increases, and the surrounding provinces decrease again. Hebei enters the HH zone from the LH zone, indicating that the sports industry is developing rapidly and that the number of bases is gradually catching up with the neighboring provinces. Guangxi entered the HL zone from the LL zone, and the number of Sports Industry Bases surpassed that of the neighboring provinces. In the economic distance weight matrix, economically developed Shanghai, Beijing, Zhejiang and Jiangsu are always located in the first quadrant. Compared with the changes between 2016 and 2021, Inner Mongolia entered the LH zone from the HH zone; i.e., the number of its own Sports Industry Bases decreased. Sichuan enters the HH zone from the LL zone, i.e., the number of bases in both the province and neighboring provinces increases. Guangxi, Hunan and Chongqing enter the LL zone from the HL zone, i.e., the number of their own bases decreases. Shandong enters the HL zone from the HH, indicating a decline in the number of bases in neighboring provinces. Hebei enters the HL zone from the LL zone, i.e., the number of bases in the province continues to rise relative to neighboring regions.

Analysis of the concentration and balance of the distribution of state-level sports industry bases

The spatial autocorrelation analysis only shows the correlation between the number of State-level Sports Industry Bases in each province and region, but it cannot further explore the degree of concentration and dispersion of their distribution, as well as the degree of equilibrium of distribution. Therefore, the spatial Gini coefficient and imbalance index are used to measure the overall distribution of State-level Sports Industry Bases, as well as the concentration and balance of distribution across the seven major geographical regions; the relevant results are shown in Table 2.

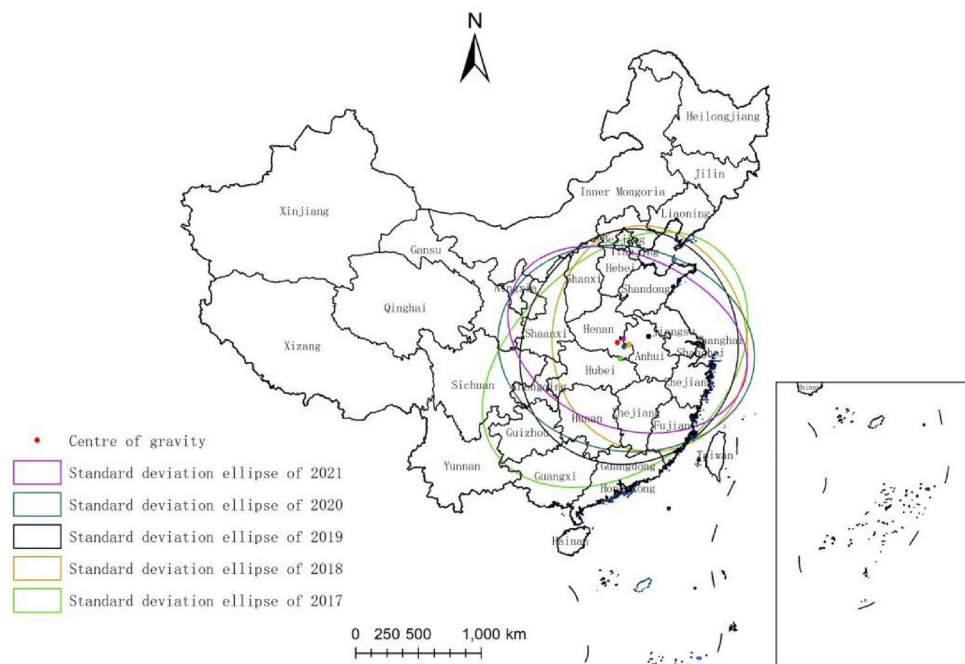
As shown in Table 2, (1) in the total distribution of State-level Sports Industry Bases in each region, East China is in the lead, followed by North China, Central China, Southwest China, South China, Northwest China and Northeast China. In terms of the average distribution of provinces and regions, East China and Central China are in the first two places, exceeding the national average. These regions are followed by North China, South China, Southwest China, Northeast China and Northwest China. (2) In terms of the spatial Gini coefficient and distribution dispersion, from a national perspective, the State-level Sports Industry Bases have the highest degree of agglomeration. From different geographical perspectives, the distribution of State-level Sports Industry Bases is most concentrated in Northeast China, followed by East China, Northwest China, South China, Central China, Southwest China, and North China, with the degree of agglomeration of State-level Sports Industry Bases decreasing in order. (3) Imbalance index: China's territory is vast, and the distribution of State-level Sports Industry Bases from the national perspective has the highest imbalance index and is the most dispersed. The distributions in East China and Northeast China are the most balanced, and the imbalance in the remaining regions ranges from low to high in Northwest China, South China, Southwest China, Central China and North China. The economic strength of the seven provinces (cities) in East China is relatively strong in the country, and the sports industry in each province and region has more distinctive characteristics, such as the sports service industry in Shanghai and the sporting goods manufacturing industry in Fujian Province. Overall, the development of State-level Sports Industry Bases in East China is relatively balanced. However, the distribution in Central China is less balanced, indicating that there are obvious provincial differences in the development of the sports industry in Shanxi, Hebei and Inner Mongolia in North China, which also causes an imbalance in the distribution of the State-level Sports Industry Bases.

Areas	Totals (Number)	Average (Number)	Imbalance index	Gini coefficient	Dispersion
Nationwide	334	10.774	0.495	0.479	0.521
Northeast China	11	3.667	0.273	0.182	0.818
North China	49	9.8	0.459	0.367	0.633
East China	174	24.857	0.261	0.223	0.777
Central China	35	11.667	0.457	0.305	0.695
South China	20	6.667	0.4	0.267	0.733
Southwest China	28	5.6	0.411	0.328	0.672
Northwest China	17	3.4	0.324	0.259	0.741

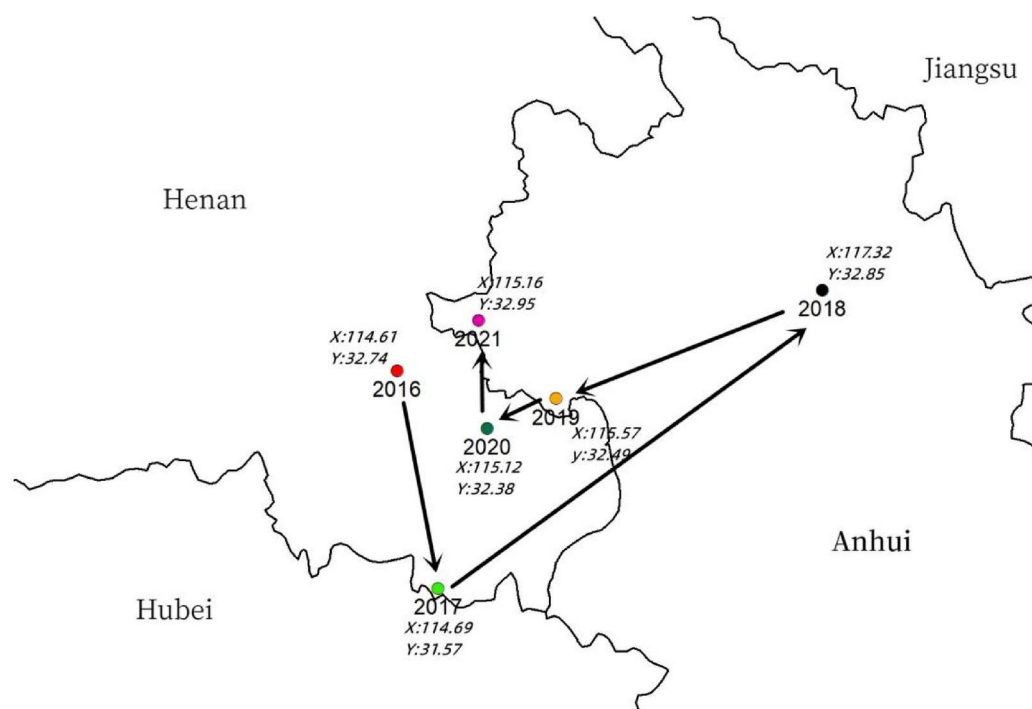
Table 2. Equilibrium indices of spatial distribution.

Standard elliptic difference analysis of state-level sports industrial bases

The distributions of the State-level Sports Industry Bases in each province and domain present different characteristics over time, and the specific transfer direction is measured by the spatial center of the gravity transfer model. As shown in Fig. 6a, the default parameter is chosen as an ellipse of 1 standard deviation size,



(a)



(b)

Fig. 6. Elliptic change of distribution standard deviation and shift of gravity (2016–2021). *Note:* This figure is based on the standard map No. GS(2020)4619 downloaded from the website of Standard Map Service of the Ministry of Natural Resources (<https://www.mnr.gov.cn/>) and the base map has not been modified. This figure was produced by using Arc Geographic Information System, ArcGIS, version 10.6 (<https://www.arcgis.com/>).

i.e., the ellipse range can encompass 68% of the Sports Industry Bases in China. In 2016, with 2049.65 km as the long half-axis (X-axis) and 1603.25 km as the short half-axis (Y-axis), the flattening rate was 446.4 km, and the direction angle was 107.86. The overall trend of the distribution is in the northwest-southeast direction. In 2017, the X-axis was extended to 2,395.93 km, the Y-axis was extended to 1661.87 km, the flattening rate was 734.06 km, and the angle of direction was 64.7. The discrete degree of distribution of the Sports Industry Bases increased, and the direction of distribution changed to southwest-northeast. In 2018, the X-axis was shortened to 1798.65 km, the Y-axis was shortened to 1548.15 km, the flatness was 250.5 km, and the angle of direction was 90.22. The distribution is "south-north", the centripetal force is increased, and the directional characteristics are weakened. In 2019, the X-axis was extended to 1875.47 km, the Y-axis was extended to 1715.42, the oblateness was 174.05, and the angle of direction was 71.82. The distribution is still "south-north", with the degree of centripetal force decreasing, the degree of dispersion increasing, and the directional characteristics further weakening. In 2020, the X-axis lengthened to 2029.89 km, the Y-axis shortened to 1628.50 km, the oblateness was 401.39, and the angle of direction was 100.39. The overall distribution is in the "northwest-southeast" direction, the centripetal force increases, and the directional characteristics increase. In 2021, the X-axis is shortened to 1955.35 km, the Y-axis is shortened to 1,367.87 km, the oblateness is 587.48, and the angle of direction is 111.12, which reflects an increase in the degree of centripetal force and a weakening of directional characteristics. In addition, when the standard ellipse parameters from 2016 and 2021 are compared, the distribution directions are both in the "southwest-northeast" direction, but the flattening rate increases in 2020, indicating that the standard ellipse has a stronger "southeast-northwest" direction. This indicates that the standard ellipse has a stronger "" orientation.

Figure 6b shows that the change in the center of gravity of the distribution of State-level Sports Industry Bases shows a slow shift, but it is mainly concentrated in the east-central provinces and regions such as Henan, Hubei and Anhui, indicating that the density of State-level Sports Industry Bases is greater in the eastern part of China, whereas the distribution in the western part is sparser. Among them, the center of gravity of the distribution in 2016–2017 shifted 153.55 km southeastward from 114.61. E, 32.74. N in southern Henan to 114.69. E, 31.57. N at the junction of Henan and Hubei and shifted 336.82 km northeastward to 117.32. E, 31.57. N in northern Anhui from 2017 to 2018. 117.32. E, 32.85. N. In 2018–2019, it moved southwestward by 200.28 km to the junction of Henan and Anhui at 115.57. E, 32.49. N. In 2019–2020, it continued to move southwesterly by 52.57 km to southern Henan at 115.12. E, 32.38 N. In 2020–2021, it moved northward by 75.79 km to 115.16 E, 32.95 N in Anhui. The counties belonging to it roughly follow Xincui County, Zhumadian city (2016)—Xin County, Xinyang city (2017)—Fengyang County, Chuzhou city (2018)—Funan County, Fuyang city (2019)—Hengchuan County, Xinyang city (2020)—Linqian County, and Fuyang city (2021) moving year by year. In the early stage of the development of the state-level Sports Industry Bases, the government's policy orientation and planning and construction focus on the eastern region. From 2018 onward, the relevant state departments gradually focused on the development of the Sports Industry Bases in the western region and further revitalized the sports industry chain in the western provinces by facilitating the flow of resources and elements, promoting the process of sports industrialization and helping the growth of the sports market.

Analysis of factors influencing the spatial distribution of state-level sports industry bases

Selection of factors

At present, research on the factors influencing the spatial distribution of State-level Sports Industry Bases is mostly qualitative and less quantitative. Therefore, to integrate the comprehensiveness and rationality of the index system, in addition to referring to other industrial bases as well as the relevant research on sports towns, another similar manifestation of sports industry agglomeration, and synthesizing the research purpose of this paper, based on data accuracy and availability, the years 2016–2021 are selected, namely, the economic strength (per capita GDP), sports consumption market capacity (per capita sports service consumption expenditure of urban households), traffic density miles of highway per 10,000 km², innovation level (number of invention patents per 10,000 people), government support (local financial expenditure on sports) and degree of urbanization development (urbanization level), to explore the effect on the number of State-level Sports Industry Bases. Some of the sports-related data statistics intersect with the education, culture and entertainment industries, drawing on Han et al.³² of data stripping ideas, the sectoral (industry) indicator weighting method is used to carry out data stripping of relevant indicators. In addition, to eliminate the influence of heteroskedasticity caused by the large range value of the constant data, the economic strength, traffic convenience, government support, sports consumption market capacity and regional innovation level are logarithmically processed, and the degree of urbanization development is not processed. Descriptive statistics for the indicators are shown in Table 3. After testing, the correlation coefficients between the indicators of each explanatory variable are less than 0.7, and the variance inflation factors (VIFs) are less than 5, indicating that there is no serious multicollinearity between the indicators, which can be further analyzed.

Regression results

The results of the panel negative binomial regression with robust standard errors, as measured by the regression model above, are shown in Table 4. In negative binomial regression, the standardized regression coefficients are generally interpreted in terms of the incidence rate ratio (IRR), or e^{β} . The effect of the independent variable can be examined in terms of the percentage change in the IRR coefficient, i.e., the percentage change in the dependent variable for each unit or standard deviation of increase in the independent variable while holding other variables constant (Table 4% column), the value obtained by this method is a standardized coefficient, which can directly measure and compare the relative importance of the impact of all independent variables on the number of State-level Sports Industry Bases³³. In Model 1, when only economic strength and government support are taken as explanatory variables, both economic strength and government support can reject the

Variable type	Variable name	Variable indicators	Variable symbol	Units	Average value	Variance	Minimum value	Maximum values
Explained Variable	Number of Sports Industry Bases	Number of Sports Industry Bases	BAS	Pcs	1.639	3.745	0	9
	Economic strength	Per capita GDP	lnPGDP	Yuan	11.010	0.166	10.186	12.123
	Government support	Local financial expenditure on sports	lnGOV	100 million Yuan	4.558	0.352	3.128	6.034
Explanatory Variable	Level of regional innovation	Patents for inventions per 10,000 persons	lnPAT	Pcs	1.826	1.140	-0.105	5.636
	Sports consumption Market volume	Per capita urban household consumption expenditure on sports services	lnEXP	Yuan	7.827	0.081	6.537	8.612
	Traffic density	Motorway miles per 10,000 km ²	lnWAY	Kilometre	5.474	1.239	0.536	7.208
	Level of urbanisation	Level of urbanisation (ratio of urban population to total population)	TOW	–	0.608	0.014	0.296	0.883

Table 3. Index Interpretation and Descriptive Statistical Analysis.

	Model 1			Model 2		
	Regression coefficient (β)	Ratio of incidence (e^{β})	%	Regression coefficient (β)	Ratio of incidence (e^{β})	%
lnPGDP	0.857*** (0.298)	2.235*** (0.702)	123.5	1.206*** (0.478)	3.339*** (1.598)	233.9
lnGOV	0.693** (0.289)	1.999** (0.578)	99.9	0.708*** (0.272)	2.029*** (0.553)	102.9
lnPAT				-0.232** (0.103)	0.792** (0.097)	-20.8
lnEXP				0.018 (0.420)	1.018 (0.428)	1.8
lnWAY				0.144 (0.148)	1.155 (0.171)	15.5
TOW				0.003 (0.074)	1.003 (0.074)	0.3
cons	-12.328*** (2.249)			-16.766*** (3.722)		
Log pseudo likelihood	-285.731		-283.670			
Pseudo R2	0.122		0.128			
Chi2	88.73		216.04			

Table 4. Regression results of panel negative binomial model. (1) Models 1 and 2 are significant at the 1% level; (2) Robust standard errors in parentheses; (3) where % is the percentage change in the number of Sports Industry Bases caused by each unit change in the independent variable when controlling for other variables; (3) Note, ***, **, * indicate that the regression results are significant at the 1%, 5%, and 10% levels, respectively, with robust standard errors in parentheses, same as the table below.

original hypothesis at the 1% significance level; i.e., they have a positive effect on the growth of the number of State-level Sports Industry Bases in this provincial region. Specifically, an increase of one unit in per capita GDP will increase the number of State-level Sports Industry Bases in this province and region by 123.5%; an increase of one unit in government financial expenditure on sports will increase the number of State-level Sports Industry Bases in this province and region by 99.9%. Model 2 adds other explanatory variables for regression, and economic strength and government support play a more obvious role in increasing the number of State-level Sports Industry Bases. Moreover, the regional innovation level has a significant negative effect on the number of State-level Sports Industry Bases. Specifically, an increase of one unit in per capita GDP will increase the number of State-level Sports Industry Bases in the province by 233.9%; an increase of one unit in government financial expenditure on sports will increase the number of bases by 102.9%; an increase of one unit in per capita invention patents will decrease the number of bases by 20.8%; in addition, the capacity of the sports consumption market, the density of traffic and the degree of urbanization do not affect the growth of the number of State-level Sports Industry Bases in the province. In addition, the sports consumption market capacity, traffic density and degree of urbanization development have no significant influence on the growth of the number of State-level Sports Industry Bases in the province.

The better the regional economic situation and the greater the government’s financial expenditure on sports is, the easier it is for the local sports industry to develop a cluster, which also leads to growth in the number of State-level Sports Industry Bases. Analyzing the reasons, the better the regional economic development is, the more people’s disposable income increases, so the sports consumption input increases. Coupled with the increase in public health awareness, enthusiasm for physical exercise and consumption has gradually increased. Moreover, public demand and the government’s advanced support of policy make the development and declaration of sports industry projects, bases and units smooth. The regional innovation level reflects local scientific and technological strength to a certain extent, and the increase in scientific and technological strength inhibits the growth of the number of Sports Industry Bases, indicating that in the development of the sports industry in various provinces and regions of China, labor, land and capital are still the main driving forces of production, whereas the factors of production, such as information and data, have not been able to efficiently enhance the efficiency of the development of sports industry agglomeration, and scientific and technological resources have

	Model 3	Model 4	Model 5	Model 6
lnPGDP	1.358*** (0.305)	1.272*** (0.333)	1.455*** (0.315)	0.804 (0.698)
lnGOV	0.351** (0.196)	0.627*** (0.149)	0.279 (0.184)	0.978*** (0.347)
lnPAT	− 0.208** (0.096)	− 0.239*** (0.098)	− 0.186* (0.099)	− 0.043 (0.296)
lnEXP	− 0.008 (0.344)	0.046 (0.341)	− 0.171 (0.332)	1.179 (0.731)
lnWAY	0.187 (0.142)	0.124 (0.156)	0.146 (0.151)	0.116 (0.301)
TOW	0.012 (0.109)	− 0.016 (0.141)	0.047 (0.060)	0.111 (0.103)
cons	− 16.778*** (3.202)	− 17.208*** (3.259)	− 16.041*** (3.279)	− 21.721*** (8.843)
Log pseudo likelihood	− 280.037	− 288.512	− 280.958	
Pseudo R2/R2		0.203		0.353
Wald chi2	106.21	128.15	109.75	41.96
Regression model	Zero-inflated Negative Binomial regression	Poisson regression	Zero-expansion Poisson regression	Mixed OLS regression

Table 5. Robustness test results of variables. (1) Models 3 to 6 are significant at the 5% level; (2) Logit models are used for the inflated factor equations in Models 3 and 5, and the variables in the models are consistent with the main equations, so the coefficients of the inflated factor models are not presented; (3) Pseudo R2 for Model 4 and R2 for Model 6.

not yet been fully exploited, which also reflects the government’s concern for the development of Sports Industry Bases and units. This finding also reflects that the government’s support for the development of Sports Industry Bases is more favorable for provinces where traditional production factors are concentrated.

Robustness test

To test the robustness of the regression results of the panel negative binomial regression model (Model 2), this paper uses the zero-inflated negative binomial regression model (Model 3), the panel Poisson regression model (Model 4), the zero-inflated Poisson regression model (Model 5), and the mixed OLS regression model (Model 6) to conduct a comparative analysis, and the results are shown in Table 5.

Notably, the estimated coefficients of Model 3–Model 5 do not indicate the marginal effect and are interpreted only as semi-elasticity, which is not numerically comparable with the estimated coefficients of Model 5; thus, only the positive and negative directions are compared. A comparison shows that the estimated coefficients of all regression models are the same, especially Model 2–Model 5, which are the same counting model and have good consistency, which verifies the conclusions in Table 4. In summary, the empirical results of this paper are robust and valid.

Conclusions, implications and recommendations
Conclusions

- (1) From 20,062,021, China named and recognized 334 State-level Sports Industry Bases (including sports industry demonstration bases, projects and units), covering 31 provinces and regions (excluding Hong Kong, Macao and Taiwan). In terms of quantity distribution, Zhejiang Province, Jiangsu Province, Fujian Province, Shanghai Municipality and Shandong Province ranked in the top 5. The introduction of the Notice on Strengthening the Construction of State-level Sports Industry Bases in 2016 expanded and clarified the concept and content system of State-level Sports Industry Bases.
- (2) The global spatial autocorrelation results reveal that under the adjacency matrix, geographic distance matrix and economic distance matrix, the provincial number of State-level Sports Industry Bases from 2016 to 2018 shows an overall increasing trend of agglomeration. In 2019, the State-level Sports Industry Bases showed a random distribution. From 2020 to 2021, under the adjacency matrix and geographic distance matrix, the spatial agglomeration status of State-level Sports Industry Bases weakened annually. In 2021, the spatial agglomeration state of State-level Sports Industry Bases decreases annually under the neighborhood matrix and geographic distance matrix. In the local spatial autocorrelation test, under the three matrices, the quadrant area of the number of State-level Sports Industry Bases in each province and domain in 2016 and 2021 exhibited a large change, i.e., the phenomenon of uneven development was more prominent.
- (3) From the perspective of the imbalance index and Gini coefficient, the distribution of State-level Sports Industry Bases in Northeast China is the most balanced, and the distribution in East China is the most concentrated. The standard ellipse difference in distribution from 20,162,021 shows that the distribution of State-level Sports Industry Bases in the past six years shows an overall trend of "discrete-Centripetal-Discrete-Centripetal-Centripetal". The center of gravity of distribution is hovering in southern Henan Province and northern Anhui Province, showing a tendency to move "east first and then west". The center of gravity of the distribution hovers in southern Henan Province and northern Anhui Province, showing a tendency of "first east then west".
- (4) Regional economic strength and government support significantly positively contribute to the growth of the number of State-level Sports Industry Bases in each province, whereas improvement in the regional innovation level decreases the number of State-level Sports Industry Bases. In addition, the capacity of the

sports consumption market, traffic density and degree of urbanization development have no significant effect on the increase in the number of State-level Sports Industry Bases.

Inspirations

- (1) An exploration of the spatial autocorrelation phenomenon of State-level Sports Industry Bases, reveals that the distribution of State-level Sports Industry Bases was uneven from 2016 to 2021 under the neighborhood matrix, geographic distance matrix and economic distance matrix. However, from 2020 onward, the agglomeration state of State-level Sports Industry Bases based on geographic location gradually weakened, and it could not be confirmed that the distribution of agglomeration could be caused by economic correlation. This extends the "geographic differentiation characteristics" proposed in previous research and triggers new thinking, which needs to explore further the factors that trigger the phenomenon of accumulation and can be extended by expanding the type of spatial matrix to explore whether the agglomeration state of the State-level Sports Industry Bases is also influenced by other resource factors. In addition, the trend of base distribution change is presented through the standard ellipse difference, which shows that since 2018, the state has gradually increased support for the sports industry in the western region and has overcome the deficiencies of the underdeveloped sports market in the west by supporting the construction of State-level Sports Industry Bases in the west and other government interventions. The unique human and natural resources in the western region can be further developed to improve the current situation of a single structure of the sports industry in the western region to increase industrial concentration.
- (2) A comparison of the factors influencing the distribution of State-level Sports Industry Bases reveals that the identification and construction of the State-level Sports Industry Bases still need to be promoted through government intervention based on a more mature market economic environment, in which the government's intervention includes central government funds, tax incentives, land concessions and preferential policies for talent, etc. Technological innovation can drive the development of Sports Industry Bases, but this study has found that the level of innovation will not increase the number of State-level Sports Industry Bases in this provincial region. To analyze the reason, the improvement in the innovation level, which is indicated mainly by patents, requires continuous investment in scientific research funds, which may impose a certain burden on the financing of local governments. For example, the State-level Sports Industry Bases in Shandong, Leling, had a research expenditure of 250 million yuan in 2015, and the State-level Sports Industry Bases in Shenzhen also reached 180 million yuan. For high investment in scientific research and innovation activities, the government and other relevant units, instead of spreading their firepower to support the development of the regional industrial base, should concentrate their efforts on doing big things, expanding the scale and scientific and technological strength of the existing sports industrial base, and driving the sustainable development of the related industrial chain. In addition, the indicators of sports consumption market capacity, traffic density and degree of urbanization development are more related to social development and market law, which are different from the government-led construction of State-level Sports Industry Bases, and although they have a weak positive effect on improving the number of Sports Industry Bases, they cannot have a significant impact.

Recommendations

Since the promulgation of the Notice on Strengthening the Construction of State-level Sports Industry Bases in 2016, State-level Sports Industry Bases have ushered in standardized development and are still in the promotion stage of development, with an uneven geographical distribution. Provinces, autonomous regions and municipalities directly under the central government are more enthusiastic about the planning and application of Sports Industry Bases but still need to improve the scientific planning capacity of construction and development. From the empirical indicators, the growth of provincial economic strength and the increase in government financial investment in sports can effectively drive the development of Sports Industry Bases in the region. On this basis, it is necessary to strengthen the communication and cooperation between city clusters to avoid the development mode of fighting alone, and regional integration development is currently the main development direction. Relying on the natural, climatic, geographical and industrial advantages of the region, developing and building unique sports projects, consolidating the foundation of the sports industry, and reasonably planning the sports industry units and projects. The government should also provide reasonable support policies for sports industry enterprises, take the market economy as the main body, cultivate the market to participate in the main consumer, give full play to the government's publicity attributes, strengthen the government's ability to attract investment, and closely connect the development of the Sports Industry Bases with the development of the economy, transportation, sports culture and urbanization of each city cluster and each province and region. Taking unique sporting events, novel industrial creativity and mature sports enterprises as the main force and using Sports Industry Bases as the benchmark, we promote the development of the sports industry in different regions of China with high quality and characteristics.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Received: 10 January 2025; Accepted: 12 March 2025

Published online: 21 March 2025

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Funding

This work was supported by the National Social Science Fund of China (Grant No. 22ATY002).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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